Lou Baya OULD ROUIS

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Research interests: White dwarfs stars (1799), Exoplanets (498), Stellar evolution (1599), Stellar mergers (2157), Stellar kinematics (1608), Magnetic fields (994), Debris disks (363), Galactic archaeology (2178)

Education

Boston University Advisor: Dr. JJ Hermes	Ph.D. Candidate in Astronomy M.A. Astronomy	2021 – Present May 2024
University of California, Los Angeles	B.S. in Physics	2017 – 2021

Research Appointments

Boston University Advisor: Dr. JJ Hermes	**Ph.D. Candidate** Conducting spectroscopic surveys (from optical to ultraviolet) to constrain how populations of massive white dwarfs relate to planetary systems evolution and stellar merger remnants.	2021-Present
UCLA Advisors: Dr. Ben Zuckerman and Dr. Beth Klein	**Undergraduate Research Assistant** Studied young forming planetary systems in the Alpha Per cluster with infrared excess emissions, trained in spectrum reduction.	2018 – 2021
UCLA Advisors: Dr. Troy Carter	**Upsilon Lab Member** Worked in research bootcamp on a Rasberry Pi high-speed camera	2018

Honors and Awards

Boston University	**NASA FINESST in Astrophysics** Proposal title: *"Missing Planets around Massive White Dwarfs?"* Awarded $150,000 of total funding as Future Investigators in NASA Earth and Space Science and Technology within ROSES-23.	2024
Boston University	**Outstanding Teaching Fellow** Awarded from the Graduate School of Arts & Sciences	2022

Awarded Grants

HST Proposal Cycle 34	7 Orbits for *"FUV Spectroscopy of Transiting Exoplanetary Debris Orbiting near the Roche Limit of a White Dwarf"*, **Co-I**	2026
Massachusetts Space Grant Fellowship	*"Inclusive Astronomy at Boston University"* $7,216 to support DEI and outreach innitiatives, **Co-I**	2024
HST Proposal Cycle 32	67 Orbits for *"The Frequency and Bulk Composition of Terrestrial Exoplanets Across Galactic History"*, **Co-I**	2024

Publications

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First Author (3)

***Searching for Massive Remnant Planetary System Hosts with White Dwarf Gaia XP Spectra**
Ould Rouis, Lou Baya; Hermes, J.J.; Bergeron, Pierre; Blouin, Simon; Arseneau, Stefan; Cunningham, Tim; Guidry, Joseph A.; Coston, Ciarra
The Astrophysical Journal, under review 2026

***White Dwarf Merger Remnants with Cooling Delays on the Q Branch Lack Strong Magnetism**
Ould Rouis, Lou Baya; Hermes, J.J.; Guidry, Joseph A.; Cheng, Sihao; Kilic, Mukremin; Vincent, Olivier; Bergeron, Pierre; Blouin, Simon; Moss, Adam; Lopez; Isaac D.; Jewett, Gracyn
The Astrophysical Journal, Volume 999, Issue 1, id. 146 2026

***Constraints on Remnant Planetary Systems as a Function of Main-sequence Mass with HST/COS**
Ould Rouis, Lou Baya; Hermes, J. J.; Gänsicke, Boris T.; Sahu, Snehalata; Koester, Detlev; Tremblay, P.-E.; Veras, Dimitri; Farihi, Jay; Heintz, Tyler M.; Gentile Fusillo, Nicola Pietro; Redfield, Seth
The Astrophysical Journal, Volume 976, Issue 2, id.156 2024

Contributing Author (5)

Now you see me, now you don't. Deep, long-lived obscuration events of white dwarfs by dust and debris from disintegrating planetesimals
van Roestel, Jan; Bhattacharjee, Soumyadeep; Guidry, Joseph A. et al. including **Ould Rouis, Lou Baya**
Astronomy & Astrophysics, submitted 2026

Near-Ultraviolet and Optical Observations of the White Dwarf ZTF J0328−1219 Reveal Variable Periodicity in its Transiting Debris
Guidry, Joseph A.; Hermes, J. J.; Vanderbosch, Zachary P. et al. including **Ould Rouis, Lou Baya**
The Astrophysical Journal, under review 2026

Transiting Planetary Debris near the Roche Limit of a White Dwarf on a 4.97 hr Orbit – and its Vanishing
Guidry, Joseph A.; Vanderbosch, Zachary P.; Hermes, J. J. et al. including **Ould Rouis, Lou Baya**
The Astrophysical Journal, Volume 992, Issue 2, id.167 2025

Using 3.4 μm Variability toward White Dwarfs as a Signpost of Remnant Planetary Systems
Guidry, Joseph A.; Hermes, J. J.; De, Kishalay; **Ould Rouis, Lou Baya** et al.
The Astronomical Journal, Volume 972, Issue 1, id.126 2024

A Test of Spectroscopic Age Estimates of White Dwarfs Using Wide WD+WD Binaries
Heintz, Tyler M.; Hermes, J. J.; Tremblay, P.-E.; **Ould Rouis, Lou Baya** et al.
The Astronomical Journal, Volume 969, Issue 1, id.68 2024

Non-Refereed Publications (3)

White dwarf planetary systems in the ultraviolet
Williams, Jamie; Bonsor, Amy; Gänsicke, Boris; Guidry, Joseph; Hermes, JJ; **Ould Rouis, Lou Baya**; Rogers, Laura; Tremblay, Pier-Emmanuel; Sahu, Snehalata; Swan, Andrew; Wilson, David; Xu, Siyi
White Paper for "Building a Roadmap for Hubble Science into the 2030s" 2026

From Hubble to HWO: Bridging the Frontier of White Dwarf Exoplanet Science
Rogers, Laura K.; Xu, Siyi; et al. including **Ould Rouis, Lou Baya**
White Paper for "Building a Roadmap for Hubble Science into the 2030s" 2026

Picture an Astronomer: Best Practices for Retaining Talent in Astrophysics
Polvin, A; Whitaker, Katherine E.; Urry, C. Megan; et al. including **Ould Rouis, Lou Baya**
White Paper from Picture an Astronomer Symposium 2025

Presentations and Invited Talks*

Contributions (10), Invitations (7)

Insights on Remnant Planetary Systems and Stellar Mergers from Massive White Dwarfs
- *Astronomy Seminar, *Trinity College Dublin* Jan. 2027 (*scheduled*)
- *FLASH Astrophysics Seminar, *NoirLAB* Oct. 2026 (*scheduled*)
- *Astronomy Seminar, *The University of Warwick* July 2026

White Dwarf Merger Remnants with Cooling Delays on the Q Branch Lack Strong Magnetism
- Contributed Talk, *EUROWD 2026: 24th European Workshop on White Dwarfs* Aug. 2026

Searching for Massive Remnant Planetary System Hosts with White Dwarf Gaia XP Spectra
- Poster, *EUROWD 2026: 24th European Workshop on White Dwarfs* Aug. 2026

Constraints on Remnant Planetary Systems as a Function of Main-sequence Mass with HST/COS
- Contributed Talk, *Boston Area Planetary Science Meeting* Jan. 2026
- *Stars and Exoplanet Seminar, *Yale University* Oct. 2025
- *Tea Talk, *California Institute of Technology* Feb. 2025
- Contributed Talk (Link), *Know Thy Star, Know Thy Planet 2, IPAC Conference* Feb. 2025

Missing Planets around Massive White Dwarfs?
- Contributed Talk, *EUROWD24: 23rd European Workshop on White Dwarfs* July 2024
- Poster (Abstract), *AASTCS 10: Extreme Solar Systems V Conference* Mar. 2024
- *Astronomy Seminar, *Tufts University* Feb. 2024
- *Exoplanet Lunch Seminar, *Center for Astrophysics | Harvard & Smithsonian* Sept. 2023
- Contributed Talk (Link), *KITP Conference: White Dwarfs from Physics to Astrophysics* Nov. 2022

Kinematics of White Dwarfs With(out) Metal Pollution
- Poster, *EUROWD22: 22nd European Workshop on White Dwarfs* Aug. 2022

Young Forming Planetary Systems in the Alpha Persei Cluster
- Contributed Talk (Abstract), *APS CUWiP Conference for Undergraduate Women in Physics* Jan. 2021
- REU Presentation, *University of California, Los Angeles* Aug. 2019

Observing Programs

Surveys

Member	DELVE/MAGIC, SDSS-V
Public User	ATLAS, DESI, Gaia, Pan-STARRS, SDSS, SkyMapper, TESS, ZTF

Selected Telescope Time

1.8-m Perkins Telescope *Flagstaff, AZ*	13 nights, **Co-I** High-speed, time-series photometry with PRISM	2025-3, 2025-4, 2026-3
4.1-m SOAR Telescope *AEON Queue*	5.6 nights (56 hr.), **PI** & 2 nights (20 hr.), **Co-I** *Low-resolution spectroscopy of white dwarfs with Goodman*	2024B–2026B
2.1-m Otto Struve Telescope *McDonald Observatory, TX*	18 nights, **Co-I** High-speed, time-series photometry with ProEM	2025-2, 2026-3
4.3-m Lowell Discovery Telescope *Happy Jack, AZ*	33 nights, **PI** & 18 nights, **Co-I** *Low-resolution spectroscopy of white dwarfs with DeVeny and high-speed, time-series photometry with LMI*	2022B–2026B

Teaching Experience

Boston University *Teaching Fellow*	**AS100 - Cosmic Controversies** Primary Instructor: Professor Mendillo *Course evaluation rating: 4.79/5* **AS105 - Alien Worlds** Primary Instructor: Professor Hermes *Course evaluation rating: 4.74/5*	Spring 2022 Fall 2021
University of California, Los Angeles *Embedded Inquiry Specialist*	**Cluster 21: History of Modern Thought** Writing and research assistance	Fall 2018 – Spring 2020
University of California, Los Angeles *Library Research Assistant*	**UCLA Library** Undergraduate Writing center	Fall 2018 – Spring 2020

Relevant Skills

Programming	Python, C++, Mathematica
Libraries	Astropy, Astroquery, IRAF, PypeIt, NumPy, Matplotlib, Pandas, SciPy, Seaborn
Other	Microsoft Office, Google Workspace, Adobe Premire Pro, TOPCAT
Languages	French [*Native*], English [*Fluent*], Spanish [*Proficient*], Arabic [*Conversational*]

Outreach

KIPP Academy Mattapan *Boston, MA*	**Astronomy Classroom Visit**, *Lead* Science activity and Q&A with a 3rd grade classroom to show what astronomers do and answer burning questions about space	March 2026
Boston University *Boston, MA*	**Deaf-friendly Interactive Night**, *Volunteer* Interactive night working with interpreters to introduce astronomical concepts to audiences aged five to eighty-five	Nov. 2024
Boston University *Boston, MA*	**Astro365**, *Volunteer* Yearly interactive nights to give astronomy exposure to low-income high school students	Dec. 2023 – Present
Boston University *Boston, MA*	**Science Train**, *Co-Founder* Monthly ride to answer astronomy questions from the general public aboard the MBTA, Boston subway system Featured in Spanish language newspaper, *El Planeta* [Linked]	May 2023 – Present
Boston University *Boston, MA*	**BU Academy Interactive Night**, *Volunteer* Interactive night with Center for Space Physics and Engineering	Apr. 2022
Boston University *Boston, MA*	**NASA Downlink**, *Volunteer* STEM education day with the ISS which gathered 500 middle- and high-school students and educators from across Massachusetts	July 2022
Boston University *Boston, MA*	**Public Open Night**, *Volunteer* Weekly event to observe the night sky through 8 inch telescopes	Sept. 2021 – Present

Professional Service

dotAstro 12 Conference *New York City, NY*	**Astro Grad Congress**, *Co-Founder* Initiative to connect graduate students in Astronomy across US institutions to promote interdepartmental discourse	Sep. 2023 – Present
Boston University *Boston, MA*	**Inclusive Astronomy**, *Co-Founder* Initiative to facilitate outreach, public engagement & DEI efforts while fostering community within the Astronomy department	Aug. 2022 – Present
Boston University *Boston, MA*	**Graduate School Application Panel**, *Moderator* Info session for BU Astronomy Society undergraduate students	Oct. 2024
Boston University *Boston, MA*	**Women in Physics Mentorship**, *Mentor* Info session for BU Astronomy Society undergraduate students	Apr. 2024 – Present
Boston University *Boston, MA*	**Graduate School Experience Panel**, *Invited Panelist* Invited by the Graduate Women in Science and Engineering (GWISE)	Nov. 2022
Boston University *Boston, MA*	**HERstory Art Proposal**, *Astronomy Jury* BU ARROWS (Advance, Recruit, Retain, & Organize Women in STEM) art installation commissioned to honor Dr. Priscilla Bok	Oct. 2021
UCLA *Los Angeles, CA*	**Academic Affairs Committee**, *Physics rep.* Participation in Physics & Astronomy curriculum matters	Sept. 2020 – June 2021
UCLA *Los Angeles, CA*	**Society of Physics Students**, *Board officer* Social media and website board officer	Oct. 2019 – June 2021

Research Advising* and Mentoring

Ariyana Bonab*	Boston University → M.S. in Astrophysics at CUNY	Nov. 2023 – Dec. 2025
Shelby Cavanaugh	Boston University → Applying for Graduate School	Apr. 2024 – Dec. 2025
Filippo Nicolai*	Boston University Academy → B.S. in Physics and Astrophysics at Boston University	May 2023 – May 2025
Nicole Flors	Boston University → Ph.D. in Astronomy at Northwestern University	Nov. 2023 – May 2024

Professional Memberships

American Astronomical Society (AAS)	Student Member	Jan. 2022 – Present
American Physical Society (APS)	Student Member	Jan. 2018 – Dec.2021

Conferences and Symposium Participation

Contribution Type: Poster* (3), Talk** (6)

EUROWD 2026: 24th European White Dwarf Workshop*,**	Vienna, AT	Aug. 2026
Local Organizing Committee (LOC)		
Boston Area Planetary Science Meeting**	Boston, MA	Jan. 2026
Local Organizing Committee (LOC), Session Chair		
Picture An Astronomer Symposium	Chicago, IL	Mar. 2025
Know Thy Star, Know Thy Planet 2, IPAC Conference**	Pasadena, CA	Feb. 2025
EUROWD24: 23rd European Workshop on White Dwarfs**	Barcelona, ES	July 2024
AASTCS 10: Extreme Solar Systems V Conference*	Christchurch, NZ	Mar. 2024
dotAstronomy 12 Conference	New York City, NY	Oct. 2023
KITP: White Dwarfs from Physics to Astrophysics**	Santa Barbara, CA	Nov. 2022
EUROWD22: 22nd European Workshop on White Dwarfs*	Tubingen, DE	Aug. 2022
APS Conference for Undergraduate Women in Physics**	Virtual	Jan. 2021
APS Conference for Undergraduate Women in Physics	Irvine, CA	Jan. 2020
APS Conference for Undergraduate Women in Physics	Santa Barbara, CA	Jan. 2019
APS Conference for Undergraduate Women in Physics	Claremont, CA	Jan. 2018